EIN3241 – Human Factors & Ergonomics

Homework 3 - Solution

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Please answer the following questions in a separate document and submit it to Canvas.

Part A: Multiple Choice Questions (26 pts)

1. In ergonomics, stress and strain refer respectively to:
   a) Internal tissue change; external demand
   b) External demand; internal change in the body
   c) Muscle force; joint torque
   d) Psychological workload; physical workload

2. The adult human skeleton contains approximately how many bones?
   a) 126
   b) 156
   c) 206
   d) 256

3. Which statement correctly distinguishes ligaments and tendons?
   a) Ligaments connect muscle to bone; tendons connect bone to bone
   b) Ligaments connect bone to bone; tendons connect muscle to bone
   c) Ligaments connect muscle to muscle; tendons connect bone to bone
   d) Both ligaments and tendons connect muscle to bone

4. Compared to dynamic effort, static muscular effort is more likely to:
   a) Increase blood flow and oxygen delivery
   b) Restrict blood flow and hasten local fatigue
   c) Decrease lactic acid buildup
   d) Reduce EMG amplitude for the same force

5. Electromyography (EMG) primarily measures:
   a) The oxygen saturation within muscle tissue
   b) The electrical activity of muscles related to force production
   c) The joint reaction moments during lifting
   d) The basal metabolic rate during rest

6. During anaerobic metabolism in muscle, pyruvic acid is converted into:
   a) Acetyl-CoA, reducing ATP regeneration
   b) Water and carbon dioxide, increasing ATP output
   c) Lactic acid, associated with fatigue and pain
   d) Glycogen, restoring energy reserves

7. Total daily metabolism is commonly expressed as:
   a) Basal + Cognitive + Thermal metabolisms
   b) Basal + Digestive + Activity metabolisms
   c) Digestive + Postural + Strength metabolisms
   d) Resting + Aerobic + Anaerobic metabolisms

8. Maximal Aerobic Power (MAP) is best described as:
   a) The highest heart rate sustainable for 8 hours
   b) The maximum oxygen uptake attainable during exercise (L/min)
   c) The fraction of cardiac output devoted to muscles at rest
   d) The oxygen cost of basal metabolism only

9. According to fatigue guidelines for an 8-hour shift, sustained work above roughly what fraction of a person's MAP leads to notable fatigue?
   a) 10–20%
   b) 20–30%
   c) 30–40%
   d) 50–60%

10. Rohmert's curve indicates that static contractions below about what percentage of MVC can be sustained for very long durations?
    a) 5% MVC
    b) 10% MVC
    c) 15% MVC
    d) 25% MVC

11. In biomechanics, kinematics and kinetics are distinguished because kinematics focuses on __, while kinetics focuses on __:
    a) Forces and moments; positions and velocities
    b) Positions, velocities, accelerations; forces and moments causing motion
    c) Muscle activation; joint geometry
    d) Energy cost; metabolic pathways

12. Which plane divides the body into left and right halves, and which anatomical term means "toward the front"?
    a) Frontal; posterior
    b) Transverse; anterior
    c) Sagittal; anterior
    d) Sagittal; inferior

13. Which body region most commonly experiences the highest proportion of work-related musculoskeletal disorders (WMSDs)?
    a) Wrist
    b) Shoulder
    c) Back
    d) Neck

Part B: Short Answer (34 pts)

1. A work activity requires 1.5 kcal/min. Determine the **total metabolism** for a woman who is 35 years old, weighs 60 kg, and is 165 cm tall (10 points).

   BM: $10(60) + 6.25(165) - 5(35) - 161 = 1295.25$ kcal

   BM: $\dfrac{1295.25}{60(24)} = 0.899$ kcal/min

   AM (given): 1.5 kcal/min

   DM: $0.1(AM + BM)$

   $= 0.1(0.899 + 1.5)$

   DM = 0.24 kcal/min

   total: AM + DM + BM = 1.5 + 0.24 + 0.899 = $\boxed{2.639 \text{ kcal/min}}$

2. A 22-year-old male is chopping wood for 50 minutes. His steady-state heart rate during the task is 130 bpm.

   (a) Estimate the energy expenditure (kcal/min) for this task (5 points).

   $\text{HR}_{\text{steady}} = 130$ bpm

   from graph ≈ 2.2 L/min oxygen consumption $\Leftarrow$ range of 2.1–2.3 accepted

   energy expenditure: $(4.8)(2.2) = 10.56$ kcal/min

   (b) What is the approximate average oxygen uptake (L) for this task? (5 points)

   Avg. oxygen uptake = $(2.2)(50) = 110$ L

3. Using the energy expenditure from Q2, calculate the allowed rest period if:

   (a) a 38-year-old female (BM = 1.0 kcal/min) performs the wood-chopping task for 100 minutes. (7 points)

   $$R_T = \frac{T(k - S)}{k\,BM} \times 1.11$$

   $$= \frac{(100)(10.56 - 4)}{10.56 - 1} \times 1.11 = 76.17 \text{ min}$$

   * Age multiplier: 1.04

   $R_T = (76.17)(1.04) = 79.22$ min

   (b) a 38-year-old male (BM = 1.1 kcal/min) performs the same task for 100 minutes. (Use $S = 4$ kcal/min for females and $S = 5$ kcal/min for males; 7 points)

$$\frac{100(10.56-5)}{10.56-1.1} \times 1.11 = 65.24 \text{ min}$$

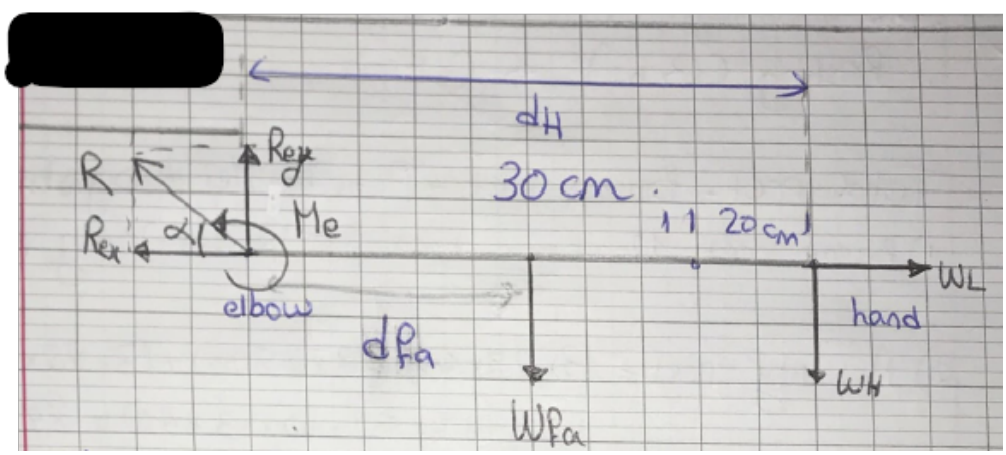
* Age multiplier: 1.04

$R_T = (65.24)(1.04) = 67.85$ min

Part C: Exercises (40 pts)

1. (10 points) Assume that a person is pulling on a handle with the forearm in the horizontal direction. Compute the reactive force and moment at the elbow (including their directions) given:

   - Pulling force = 20 kg ≈ 196 N
   - Mass of person = 80 kg
   - Hand length = 20 cm
   - Forearm length = 30 cm



$WL = 20\,\text{kg} \times 9.8\,\text{N/kg} \Rightarrow WL = 196\,\text{N}$

$WH = (0.006)(80\,\text{kg})(9.8\,\text{N/kg}) = 4.704\,\text{N}$

$W_{FA} = 0.017 \times 80\,\text{kg} \times 9.8\,\text{N/kg} = 13.328\,\text{N}$

$d_{FA} = 0.43 \times 0.3 = 0.129\,\text{m}$

$d_H = 0.3 + 0.506 \times 20/100 = 0.4012\,\text{m}$

$\Rightarrow \sum F_x = 0 \quad \Rightarrow WL = R_{ex} \quad \Rightarrow R_{ex} = WL \Rightarrow R_{ex} = 196\,\text{N}$ left

$\Rightarrow \sum F_y = 0 \quad \Rightarrow R_{ey} - W_{FA} - WH = 0$

$\Rightarrow R_{ey} = W_{FA} + WH$

$\Rightarrow R_{ey} = 13.328 + 4.704 = 18.032\,\text{N}$ upward

$R = \sqrt{196^2 + 18.032^2} = 196.8277\,\text{N}$

so R is upward left, making angle α with horizontal:

EXTRA FOR PRACTICE BUT NOT REQUIRED: $\alpha = \tan^{-1}\left(\frac{R_{ey}}{R_{ex}}\right) = \tan^{-1}\left(\frac{18.032}{196}\right) = 5.26°$

$\Rightarrow \sum M_H = 0$

$-W_{FA} \times d_{FA} - WH \times d_H + M_e = 0$

$M_e = 13.328 \times 0.129 + 4.704 \times 0.4012$

$M_e = 3.6066\,\text{N·m}$ (counterclockwise)

2. (30 points) Consider the following information for the lifting task below:

   - Mass of box = 15 kg; Mass of person = 75 kg
   - Hand length = 18 cm; Forearm length = 30 cm; Upper arm length = 34 cm
   - Assume the weight of the box is applied at the center of gravity of the hands.

   NOTE: This question was presented in the slides. In class, we did the one-segment approach. Here, please do the **Segment-to-Segment Approach** to find the reaction force and moment of the shoulder to confirm the numbers we got in class, and then continue to find the reaction force and moment on the lower back. Assume that the total length of the subject's torso, neck, and head is 75 cm.

   Along the way, prove that:

$$R_E = 90.405\,\text{N}\ (\textit{upwards})$$
$$M_E = 27.78\,\text{Nm}\ (CCW)$$

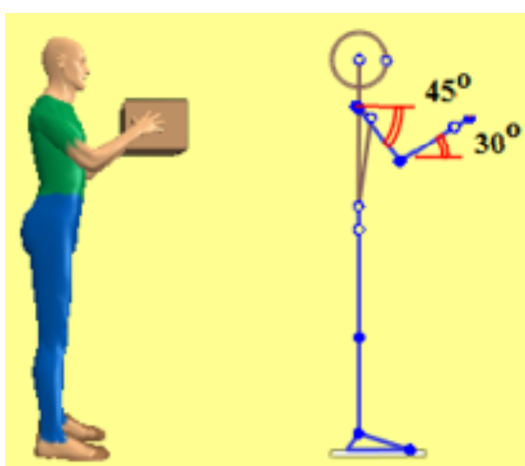


Figure 1: Lifting task setup showing the subject holding a box and the biomechanical model with arm joint angles of 45° (shoulder) and 30° (elbow).

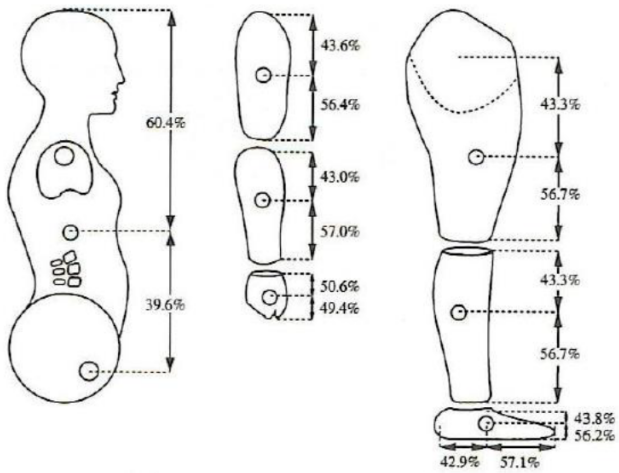
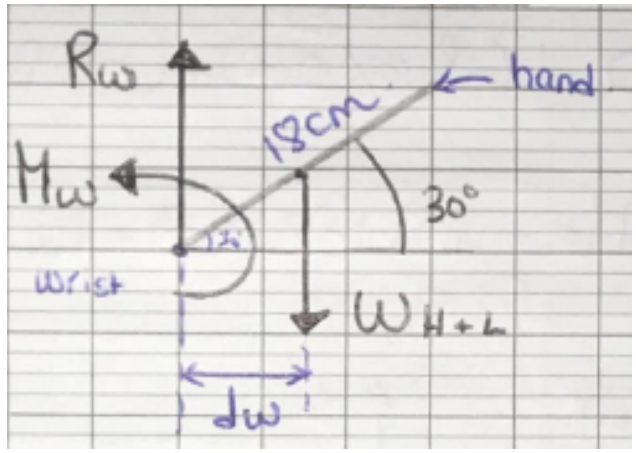


Figure 2: The locations of centers of mass in the body segments in the sagittal plane indicated by percentages of the body segments. Adapted from Dempster (1955).

Group body segments as a percentage of		Individual body segment mass as a percentage of		
	Total body mass (%)		Group segment mass (%)	Total body mass (%)
Head and neck	8.4	Head	73.8	6.2
		Neck	26.2	2.2
Torso (trunk)	50.0	Thorax	43.8	21.9
		Lumbar	29.4	14.7
		Pelvis	26.8	13.4
Each arm (total)	5.1	Upper arm	54.9	2.8
		Forearm	33.3	1.7
		Hand	11.8	0.6
Each leg (total)	15.7	Thigh	63.7	10.0
		Lower leg (shank)	27.4	4.3
		Foot	8.9	1.4

Figure 3: Masses of body segments as a percentage of the whole body mass.

Segment 1: Hand



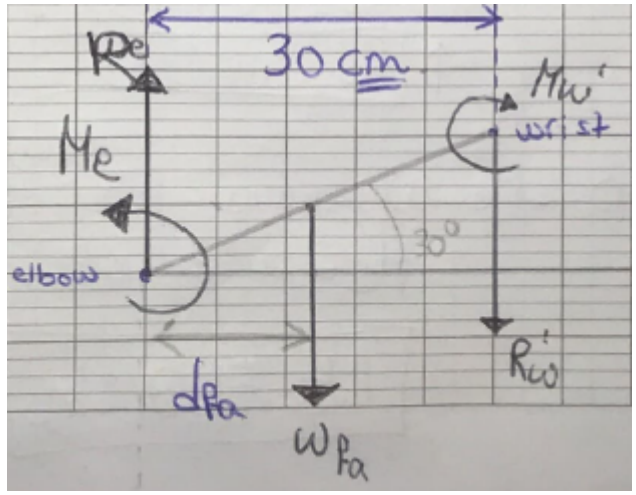
$$WL = \frac{75\,\text{kg}}{2} \times 9.8\,\text{N/kg} = 73.5\,\text{N}$$
$$WH = (0.006)(75\,\text{kg})(9.8\,\text{N/kg}) = 4.41\,\text{N}$$
$$\Rightarrow W_{H+L} = 77.91\,\text{N}$$

Point of Reference: Wrist

$$\Rightarrow \sum F_y = 0$$
$$-W_{H+L} + R_W = 0 \Rightarrow R_W = 77.91\,\text{N (upward)}$$

$$\Rightarrow \sum M_W = 0$$
$$(-W_{H+L} \times d_W) + M_W = 0$$
$$M_W = (W_{H+L}) \times d_W$$
$$M_W = 77.91 \times (\cos 30^\circ \times 0.506 \times 0.18)$$
$$M_W = 6.145\,\text{N·m (counterclockwise)}$$

Segment 2: Forearm



$W_{FA} = (0.017)(75\,\text{kg})(9.8\,\text{N/kg}) = 12.5\,\text{N}$

$d_{FA} = (0.43)(0.3) \times \cos 30° = 0.1117\,\text{m}$

Point of Reference: Elbow

$\Rightarrow \sum F_y = 0$

$-R_W + W_{FA} + R_E = 0$

$R_E = 77.91 + 12.5 = 90.41\,\text{N (upward)}$

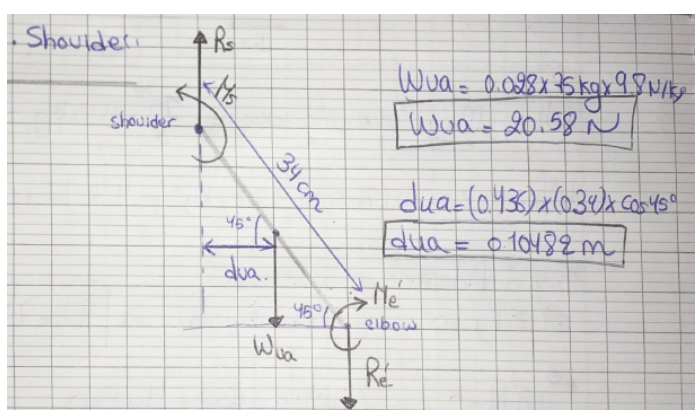
$\Rightarrow \sum M_E = 0$

$(-R_W \times d_W) + (W_{FA} \times d_{FA}) + M_E - M_W = 0$

$M_E = (77.91 \times 0.3 \times \cos 30°) + (12.5 \times 0.1117) + 6.145$

$M_E = 27.783\,\text{N·m (counterclockwise)}$

Segment 3: Upper Arm



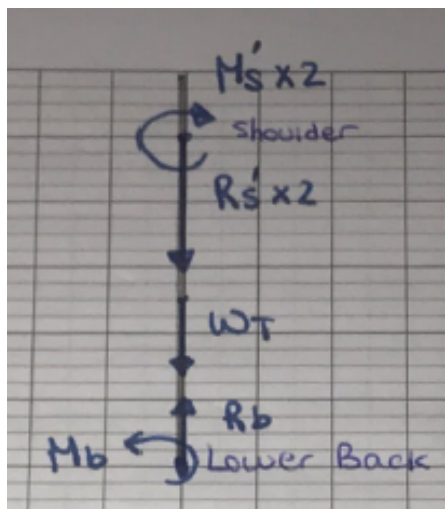
$W_{UA} = (0.028)(75\,\text{kg})(9.8\,\text{N/kg}) = 20.58\,\text{N}$
$d_{UA} = (0.43)(0.34) \times \cos 45° = 0.10482\,\text{m}$

Point of Reference: Shoulder

$\Rightarrow \sum F_y = 0$
$R'_E - W_{UA} + R_S = 0$
$R_S = R'_E + W_{UA}$
$R_S = 90.41 + 20.58 = 110.99\,\text{N}$ (upward)

$\Rightarrow \sum M_S = 0$
$-M_E - W_{UA} \times d_{UA} - R'_E \times 0.34 \times \cos 45° + M_S = 0$
$M_S = 20.58 \times 0.10482 + 90.41 \times 0.24 + 27.783$
$M_S = 51.64\,\text{N·m}$ (counterclockwise)

Segment 4: Trunk (Lower Back)



We already calculated R_S and M_S in the previous segment:
$R_S = 110.99\,\text{N}$ and $M_S = 51.68\,\text{N·m}$ (counterclockwise)

$W_T = (0.062 + 0.022 + 0.219 + 0.147) \times 75\,\text{kg} \times 9.8\,\text{N/kg}$
$W_T = 330.75\,\text{N}$

$\Rightarrow \sum F_y = 0$
$-2R_S - W_T + R_B = 0$
$R_B = 2(110.99) + 330.75 = 552.73\,\text{N}$ (upward)

$\Rightarrow \sum M_B = 0$
$-2M_S + M_B = 0$
$M_B = 2(51.68) = 103.36\,\text{N·m}$ (counterclockwise)